Vecta-DWI: Data Birth Integrity Assessment Framework

Inputs
BIDS sidecar JSON / bval / bvec
DICOM source archive (optional)
Versioned Specification
22 Variables (5 domains)
8 Criteria (dwi_connectomics)
Intended-use Profile
Assessment Engine
BIDS Collector
DICOM Collector
Variable Extractors (22 variables)
Criteria Engine (8 criteria)
Output Assembler + Schema Validator (14 JSON Schemas)
Structured Outputs
Per-session Assessment JSON (canonical)
Readiness State + Findings
Criterion Attribution + Severity
Cohort Aggregate TSV / HTML
Evidence Chain (BIDS field → variable → criterion → finding)
Formal Properties
Null safety unknown ≠ false
Deterministic (spec-version pinned)
Intended-use conditional
Profile-relative readiness
Input data
Specification (YAML)
Engine components
Structured outputs
Formal properties